PART A: PROBLEM-SOLVING QUESTIONS **[40 MARKS]**

QUESTION 1 (dr alif) **[10 MARKS]**

Identify the output of the flowchart in **Figure 1** by tracing the value of the variables. Complete the tracing table in **Table 1** to write your answers. Then, show the calculations for variable ***nd***, ***all*** and ***div*** in **Table 2**.

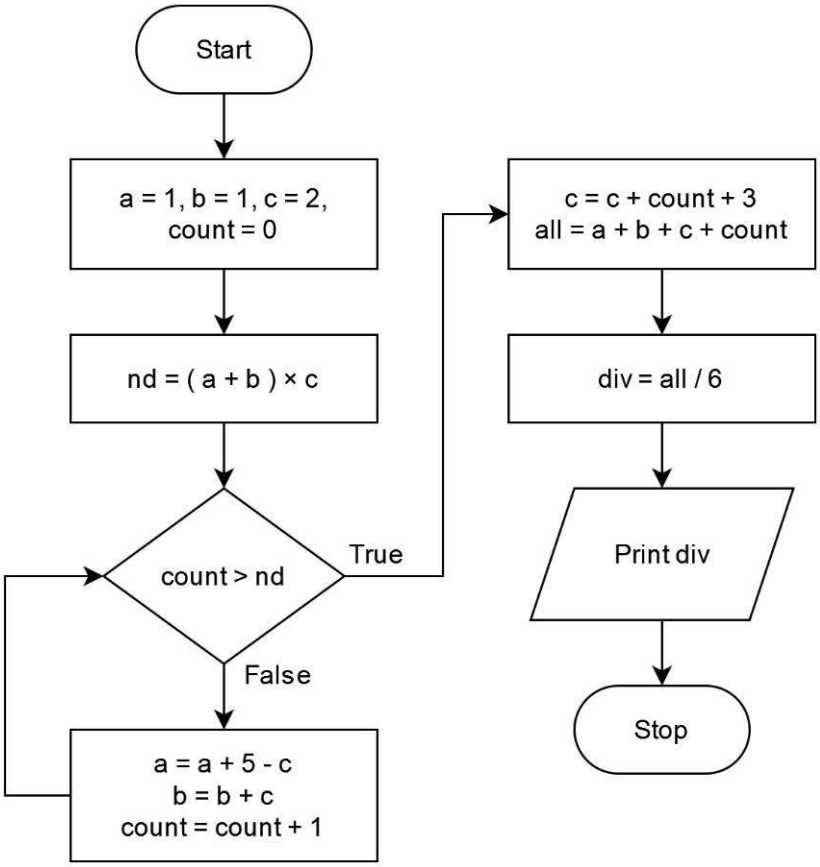


Figure 1: Flowchart for Question 1

Table 1: Tracing table for Question 1

a	b	c	count	count > nd	Output
1	***1***	***2***	***0***		

Table 2: Variables calculation for Question 1

Calculation for variable nd, all and div:

Answer Scheme

a	b	c	count	count > nd	Output
1	***1***	***2***	***0***	False	
4	3		1	False	
7	5		2	False	
10	7		3	False	
13	9		4	False	
16	11		5	True	
		10			7 **//1m**

Calculation:
nd = (1+1) × 2 = 4 **//1m**
all = 16 + 11 + 10 + 5 = 42 **//1m**
div = 42 / 6 = 7 **//1m**

0.3M each = 10 × 0.3 = **3m**
0.25M each = 12 × 0.25 = **3m**

QUESTION 2 (dr khatib) **[10 MARKS]**

Draw a **flowchart** to calculate the price of items after discount. The steps for calculating are as follows:

(a) Get the prices of items.

(b) If the total prices of the items are more than RM 50.00, call a user-defined function named *"disc10"* then calculate the total price after discount and return it to the main function.

(c) If the total prices of the items are more than RM 100.00, call a user-defined function name *"disc15"* then calculate the total price after discount and return it to the main function.

(d) Finally, in the main function, display the total number of items, the original total price before discount and the price after discount.

Table 3 shows the discount tag offer for the total price of the items.

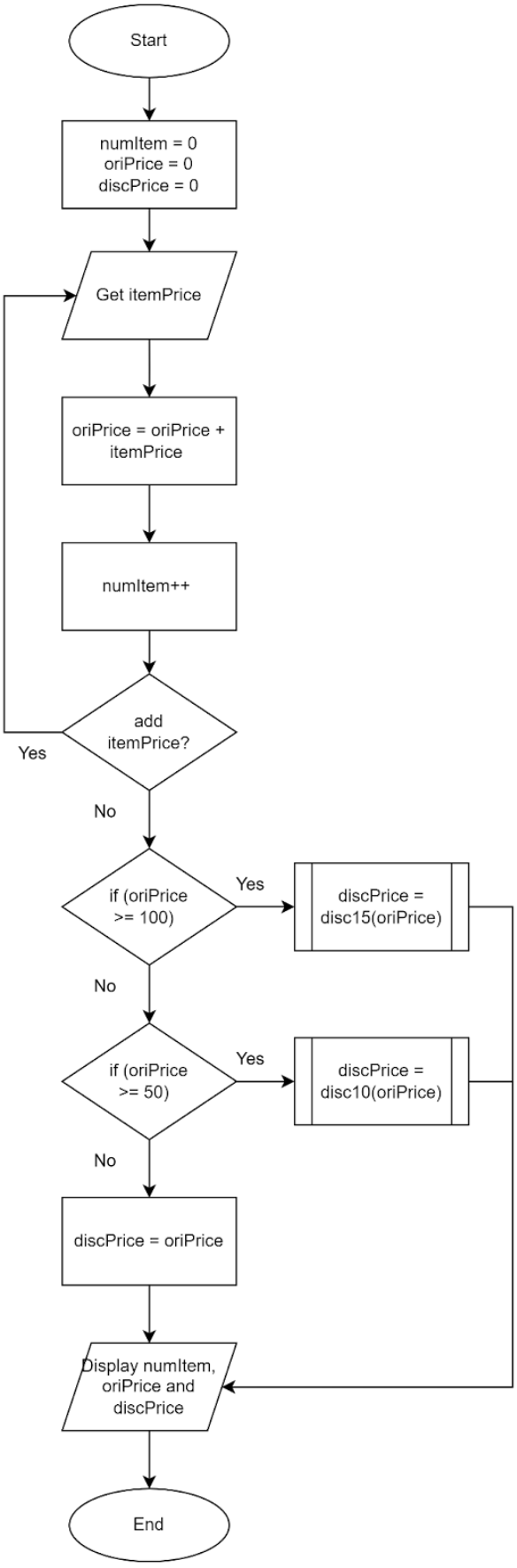
Table 3: Discount for each tag for Question 2

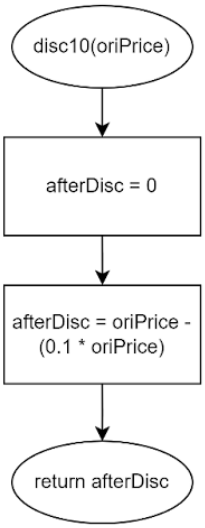
Discount Tag	Discount
disc10	10%
disc15	15%

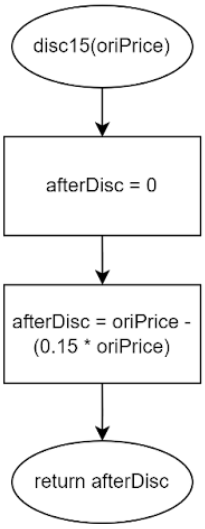
Note: User input in step (a), allows more than an item price to be added. Calculation of the total price after discount must be done in the user defined function as named in (b) and (c) accordingly. The price after discount must be returned to the main function once the calculation is finished.

Answer space for Question 2

Answer Scheme







QUESTION 3 (dr goh) **[10 MARKS]**

Please complete the flow chart given in **Figure 2** so that it will produce any outputs given in the example 1 to 3. Write your answer in **Table 4**.

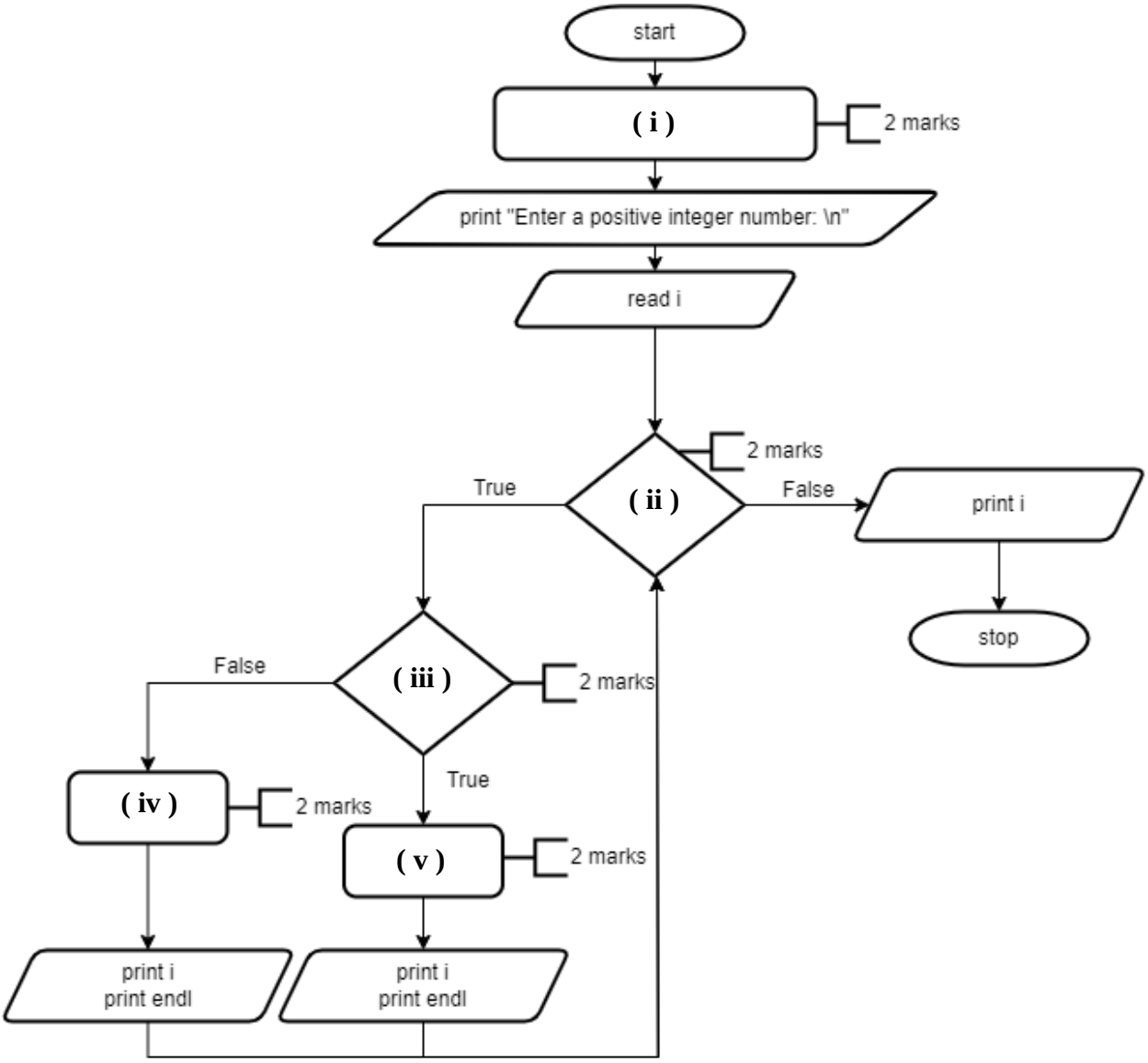


Figure 2: Flowchart for Question 3

Example output 1:

```
Enter a positive integer
number:
23 [Enter]
23
```

Example output 2:

```
Enter a positive integer
number:
14 [Enter]
15
15
```

Example output 3:

```
Enter a positive integer
number:
9 [Enter]
11
11
12
13
14
15
15
```

Note: the bolded text are the inputs from the user

Table 4: Answer for Question 3

Question	Answer
(i)	`set i=0`
(ii)	`i<15`
(iii)	`i<10`
(iv)	`i++`
(v)	`i=i+2`

QUESTION 4 (dr eiliyah) **[10 MARKS]**

Write a complete **pseudocode** which will get several inputs from the user: bank 'account balance', the 'account type' and 'account level' they have. Apply **selection** concept. Based on this information and the below rate table from Bank ABC in **Table 5**, determine the interest rate they are receiving.

Table 5: Rate table of Bank ABC for Question 4

Type of Account	Level	Minimum Balance	Interest Rate
Student	Standard	RM25	1.3 %
Personal	Standard	RM0	1.2 %
Personal	Gold	RM1000	1.9 %
Personal	Gold	RM5000	2.3 %
Business	Standard	RM1500	1.7 %
Business	Platinum	RM10000	2.5%

Answer space for Question 4

Answer Scheme

```
INPUT user account type, account level, and balance.
IF account type is “personal” and account level is “standard” and balance >=
0
   PRINT “Interest Rate is 1.2%”
      ELSEIF account type is “personal” and account level is “gold” and
      balance >= 5000
            PRINT “Interest Rate is 2.3%”
      ELSEIF account type is “personal” and account level is “standard” and
      balance >=1000
            PRINT “Interest Rate is 1.9%”
      ELSEIF account type is “business” and account level is “standard” and
      balance >=1500
            PRINT “Interest Rate is 1.7%”
      ELSEIF account type is “business” and account level is “gold” and
      balance >= 10000
            PRINT “Interest Rate is 2.5%”
ELSE
      PRINT “Error: the account”
      ENDIF
```

PART B: PROGRAMMING QUESTIONS **[60 MARKS]**

QUESTION 1 **[15 MARKS]**

(a) **Figure 1** shows the flowchart that asks the user to calculate the total mark of a test. The C++ code of the flowchart is shown in **Figure 2**. You are required to complete the C++ code in **Figure 2** and write your answer in **Table 1**. (dr khatib) [5 marks]

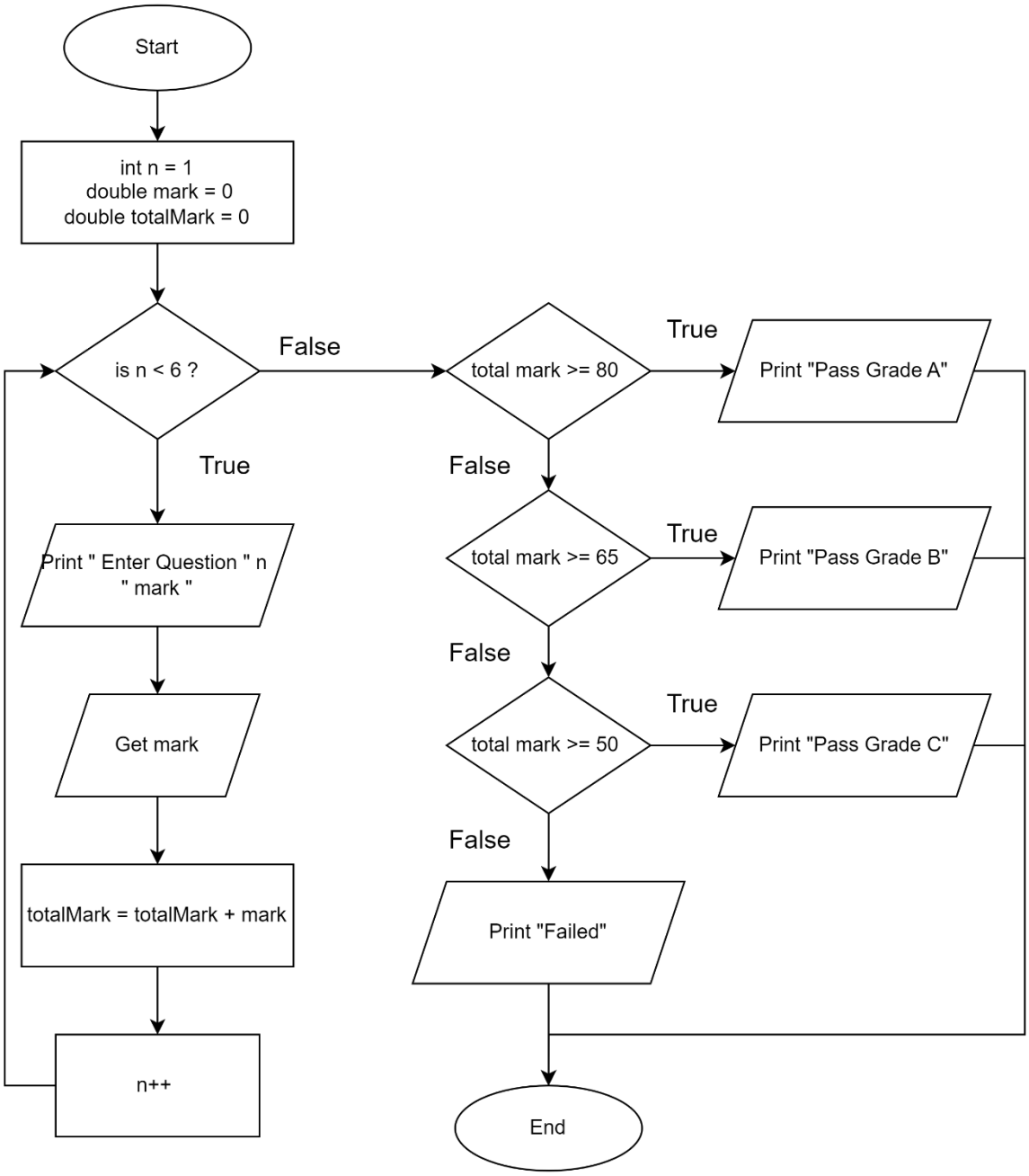


Figure 1: Flowchart for Question 1(a)

```
#include <iostream>
```

```
using namespace std;

int main() {

    int n = 1;
    double mark, totalMark = 0;

    _____(i)_____
    {
        cout << "Enter Question"<< n <<" mark ";
        cin >> mark;

        _____(ii)_____

        n++;
    }

    _____(iii)_____
        cout << "Pass Grade A";

    _____(iv)_____
        cout << "Pass Grade B";

    _____(v)_____
        cout << "Pass Grade C";

    else
        cout << "Failed";

    return 0;
}
```

Figure 2: C++ program code for Question 1(a)

Table 1: Answer for Question 1(a)

Question	Answer
(i)	`while (n<6)`
(ii)	`totalMark = totalMark + mark;`
(iii)	`if (totalMark >= 80)`
(iv)	`else if (totalMark >= 65)`
(v)	`else if (totalMark >= 50)`

(b) From the C++ program code in **Figure 3**, trace the output and write your answer in the answer column. (dr goh) [5 marks]

Line	Code	Answer
1	`#include <iostream>`	
2	`using namespace std;`	
3	`int main(){`	
4	`    int a, b = 2, c, d = 1, e=0;`	
5	`    float p = 3.0, q=0.0;`	
6	`    char x='1', y='A', z='?';`	
7	`    bool m=0;`	
8	`    if (b>m)`	
9	`        m=b;`	
10	`    else`	
11	`        m=d;`	
12		
13	`    if (b>e)`	
14	`        e=e+2;`	
15	`    else`	
16	`        a=b;`	
17	`    q = static_cast<float>(e/5) * d;`	
18	`    p = static_cast <float>(b*d--);`	
19	`    c = ++a / 2 + e;`	
20		
21	`    if (x==1)`	
22	`        m=1;`	
23	`    else`	
24	`        y='D';`	
26	**`    cout << a << endl;`**	1
27	**`    cout << b << endl;`**	2
28	**`    cout << p << endl;`**	2
29	**`    cout << q << endl;`**	0
30	**`    cout << y << endl;`**	D
31	`    return 0;`	
32	`}`	

Figure 3: C++ program code for Question 1(b)

(c) Identify the output for the program segment shown in **Table 2**. Fill in the **Output** column of the table with your answers. (dr alif) [5 marks]

Table 2: Program segment and output for Question 1(c)

Line	Code	Output
1	`int x = 10, y = -3;`	
2	`bool k = true;`	
3		
4	`cout << ( (x == 10) || (x < y) ) << endl;`	1
5	`cout << ( (x = y) && (k = 0) ) << endl;`	0
6	`cout << ( (y == -3) && (x == 10) ) << endl;`	0
7	`cout << ( (y != x) || (k != true) ) << endl;`	1
8	`cout << ( (k == true) && (x >= 10) ) << endl;`	0

QUESTION 2 (dr suhaimie) **[15 MARKS]**

Figure 4 below shows the complete C++ code.

```
1  #include <iostream>
2  using namespace std;
3
4  int main()
5  {
6     float choice;
7     int counter;
8     bool flag = 0;
9
10    cout << "\tEnter choice   :  ";
11    cin >> choice;
12    cout << "\tEnter counter  :  ";
13    cin >> counter;
14
15    while (!flag && (counter % 2)==0;)
16    {
17       if ((choice != 'a') && (choice != 'b') &&
18       (choice != 'c'))
19       {  counter += 3;
20          flag = 0;
21       else
22          flag = 1;
23          counter *= 3;}
24    }
25
26    for (int i = 0; i < counter;)
27    {
28       switch ("choice")
29       {
```

```
30              case 'a' : ++i; cout << "one\n";
31                         break;
32              case 'b' : i+=2; cout << "two\n";
33                         break;
34              case 'c' : i+=3; cout << "three\n";
35              default  : i+=4; cout << "four\n";
36                         break;
37          }
38      }
39      return 0;
40  }
```

Figure 4: C++ program code for Question 2

(a) From the **Figure 4**, there are 5 identified syntax errors in the code in the usage of the control structure codes (repetition and selection). Find the line of the code and write the correct code. Write your answer in **Table 3**. [5 marks]

Table 3: Answer for Question 2(a)

Line No.	Correction
6	Declare as a char data type / `float choice`
15	Remove the ";" / **`while`** `(!flag && (counter % 2)==0)`
20	Add close braces "}" / `flag = 0; }`
21	Add open braces "{" / `{ flag = 1;`
28	Remove the double quote in the choice variable / **`switch`** `(choice)`

(b) Assuming that you have identified and found all the errors in Question 2(a) above, trace the program by filling in the values of the variables in the **Table 4** below when the initial value is as shown below. [10 marks]

Table 4: Answer for Question 2(b)

flag	choice	counter
0	c	4
1		12

i	On display
0	
3	three
7	four
10	three
14	four

flag	choice	counter
0	d	12
0		15

i	On display
0	
4	four
8	four
12	four
16	four

QUESTION 3 (dr jamilah) **[15 MARKS]**

(a) **Figure 5** below shows the incomplete C++ code for calculating area for a house, complete the C++ code for the program and write your answer in **Table 5**. [5 marks]

```
#include <iostream>
using namespace std;

int main() {
    double length, width;

    //declares a character array named typeOfHouse capable of
    holding 50 characters.
    _____(i);

    int numOfTimes,
    count = 0;

    cout << "Enter the type of house: ";
    _____(ii)>> typeOfHouse;

    cout<<"How many times do you want to calculate the area? ";
    cin >> numOfTimes;

    while(_____(iii))
    {
        cout << "Enter length and width of the rectangle: ";
        cin >> length >> width;

        double area = length * width;

        cout << "Area of the rectangle (" << length << " x "
        << width << ") for house type '" << typeOfHouse << "'
        is: ";

        //convert data type for the area to int
        cout << _____(iv) << endl;

        //Updating of counter value (increase by 1) during each
        iteration
```

```
        ____________(v);
    }

    return 0;
}
```

Figure 5: C++ program code for Question 3(a)

Table 5: Answer for Question 3(a)

Question	Answer
(i)	`char typeOfHouse[50];`
(ii)	`cin`
(iii)	`count < numOfTimes`
(iv)	`static_cast<int>(area)`
(v)	`count++`

(b) **Figure 6** show a C++ program code that prompts the user to input two numbers that are **`multiplicationNum`** for the multiplication table and **`divisionNum`** for the division table. Additionally, the user should input a range limit (**`limit`**). The program displays the output of the multiplication table for **`multiplicationNum`** and the division table for **`divisionNum`** from 1 up to the specified **`limit`**. **Figure 7** shows an example of the expected output for the specified value of variables below. [10 marks]

From **Figure 6** and **7**, answer the following questions:

(i). Write code segment for multiplication table. (5 marks)

(ii). Write code segment for division table. Use **static cast** to convert the output to **double** number. (5 marks)

```
#include <iostream>
using namespace std;

int main() {
    int multiplicationNum, divisionNum, limit;

    cout << "Enter a number for multiplication table: ";
    cin >> multiplicationNum;

    cout << "Enter a number for division table: ";
    cin >> divisionNum;
```

```
    cout << "Enter the limit: ";
    cin >> limit;

cout << "Multiplication table for " << multiplicationNum << "
up to " << limit << ":" << endl;

    // (i) Write code segment for multiplication table.


cout << "Division table for " << divisionNum << " up to " <<
limit << ":" << endl;

    // (ii) Write code segment for division table.

    return 0;
}
```

Figure 6: C++ program code for Question 3(b)

```
Enter a number for multiplication table: 10 [Enter]
Enter a number for division table: 12 [Enter]
Enter the limit: 5 [Enter]

Multiplication table for 10 up to 5:
10 x 1 = 10
10 x 2 = 20
10 x 3 = 30
10 x 4 = 40
10 x 5 = 50

Division table for 12 up to 5:
12 / 1 = 12
12 / 2 = 6
12 / 3 = 4
12 / 4 = 3
12 / 5 = 2.4
```

Figure 7: Example of the C++ program expected output for Question 3(b)

Answer space for Question (i) and (ii)

ANSWER

i) /Code segment for multiplication table - 5 marks for correct answer

```
for (int i = 1; i <= limit; ++i) {
     cout << multiplicationNum << " x " << i << " = " <<
     (multiplicationNum * i) << endl;
}
```

ii) //Code segment for division table - 5 marks for correct answer

```
for (int i = 1; i <= limit; ++i) {
     double result = static_cast<double>(divisionNum) / i;
     cout << divisionNum << " / " << i << " = " << result <<
     endl;
}
```

QUESTION 4 (dr nies) **[15 MARKS]**

(a) From the C++ program code in **Figure 8**, trace the output and write your answer in the answer column. [5 marks]

Line	Code	Answer
1	`#include <iostream>`	
2	`using namespace std;`	
3		
4	`int main() {`	
5	`int a = 0, b = 0, c = 0, d = 0;`	
6		
7	`cout << "\t\t" << "a" << "\t" << "b" << "\t" << "c" << "\t" << "d\n";`	
8		
9	`cout << "initialize" << "\t";`	
10		
11	`cout << a << "\t" << b << "\t" << c << "\t" << d << "\n";`	
12		1
13	**`(a+=2)*(--a);`**	0

14	`(--((++b)%=3));`	0.2
15	`((++c)/5.0);`	3
16	`++(d+=2);`	
17		
18	`cout << "update" << "\t\t";`	
19		1 0 1 3
20	`cout << a << "\t" << b << "\t" << c << "\t" << d << "\n";`	
21		
22	`return 0;`	
23	`}`	

Figure 8: C++ program code for Question 4(a)

(b) From the C++ program code in **Figure 9**, answer the question (i), (ii) and (iii) by writing the code segments for each task to produce the output in **Figure 10**. [10 marks]

(i). Write the code segment to print even numbers from 1 to 10 (3 marks)

(ii). Write the code segment to print odd numbers from 1 to 10 (3 marks)

(iii). Write the code segment to print prime numbers from 1 to 10 (4 marks)

Note: You are compulsory to apply a **LOOP** using **for/while/do…while AND** apply a **SELECTION** using **if/if…else/switch…case**.

```
#include <iostream>
using namespace std;
int main(){
     cout << "List numbers from 1 to 10\n";

     // Print numbers from 1 to 10
     for (int i = 1; i <= 10; ++i) {

          cout << i << " ";

          cout << "\n\nEven numbers from 1 to 10\n";
          // (i) Write the code segment to print even numbers
          from 1 to 10

          cout << "\n\nOdd numbers from 1 to 10\n";
          // (ii) Write the code segment to print odd numbers
          from 1 to 10
```

```
        cout << "\n\nPrime numbers from 1 to 10\n";
        // (iii) Write the code segment to print prime numbers
        from 1 to 10
    }
    return 0;
}
```

Figure 9: C++ program code for Question 4(b)

```
List numbers from 1 to 10
1 2 3 4 5 6 7 8 9 10

Even numbers from 1 to 10
2 4 6 8 10

Odd numbers from 1 to 10
1 3 5 7 9

Prime numbers from 1 to 10
2 3 5 7
```

Figure 10: C++ program output for Question 4(b)

Answer space for Question (i), (ii) and (ii)

ANSWER

(i)

Ans1

```
for (int i = 1; i <= 10; ++i) {
    // Check if the number is even
    if (i % 2 == 0) cout << i << " ";
}
```

Ans2

```
for (int i = 2; i <= 10; i += 2) {
    cout << i << " ";
}
```

(ii)

Ans1

```
for (int i = 1; i <= 10; ++i) {
    // Check if the number is even
    if (i % 2 == 1) cout << i << " ";
}
```

Ans2

```
for (int i = 1; i <= 10; i += 2) {
    cout << i << " ";
}
```

(iii)

Ans1

```
for (int i = 2; i <= 10; ++i) {
    int j;
    for (j = 2; j * j <= i; ++j) {
        if (i % j == 0) break;
    }
    if (j * j > i) cout << i << " ";
}
```

Ans1

```
for (int i = 1; i <= 10; ++i) {
    bool isPrime = true;
    if (i <= 1) isPrime = false;
      else {
        for (int j = 2; j * j <= i; ++j) {
            if (i % j == 0) {
                isPrime = false;
                break;
            }
        }
    }
    if (isPrime) cout << i << " ";
}
```

End of Question